

SOFTWARE ARCHITECTURE

The Software Architecture of the Monkey Head Project defines how Huey's AI/OS and supporting systems are structured, deployed, and managed. It emphasizes modularity, scalability, and resilience to ensure long-term adaptability.

Key Elements:

- OS Tiers – Layered approach using MacroOS, MicroOS, and NanoOS for strategic, operational, and real-time control.
- Orchestration – Container-based workloads managed by Kubernetes for automated scaling, deployment, and recovery.
- Storage Models – Includes the Bees & Honey biomimetic storage approach for efficiency and redundancy.
- Specialized Protocols – Unique frameworks like Plane & Submarine Logistics for system safety and redundancy under extreme conditions.

Design Principles:

- Modularity – Each functional unit runs independently, allowing upgrades without impacting the whole.
- Scalability – New services or capabilities can be added with minimal reconfiguration.
- Resilience – Built-in fault tolerance and recovery mechanisms maintain stability under stress.

Governance Integration:

- All software changes are documented in the Unified Audit Log.
- Governance approval ensures compliance with ethical and operational standards.

The following pages will break down each software element in detail, including operating system tiers, orchestration, storage systems, and safety protocols.